

TIMEDEXEC – User Commands Manual

NAME

timedexec - execute a program with strict CPU time and memory limits

SYNOPSIS

```
./timedexec [-t <cpu_time_seconds>] [-m <memory_limit_bytes>] command [args...]
```

DESCRIPTION

timedexec acts as a resource supervisor for any specified command. It runs the target program in a child process and enforces hard limits on the amount of CPU time and virtual memory the program is allowed to consume.

The program uses `setrlimit()` to interact directly with the Linux kernel, ensuring that if the limits are exceeded, the kernel automatically terminates the process.

OPTIONS

-t <seconds> : Specifies the maximum CPU time (in seconds) the command is allowed to use.
-m <bytes> : Specifies the maximum virtual memory (in bytes) the command is allowed to allocate.

EXAMPLES

```
./timedexec -t 2 -m 500000000 ./my_program  
./timedexec -t 5 ls -l
```

SYSTEM CALLS USED

`fork()` - creates a child process to run the target command
`execvp()` - executes the target command inside the child process
`setrlimit()` - sets the `RLIMIT_CPU` and `RLIMIT_AS` resource limits
`waitpid()` - allows the parent process to wait for the child and read its exit status
`getopt()` - parses command-line arguments

ERROR HANDLING

Handles missing commands/arguments gracefully.
Reports if `fork()`, `execvp()`, or `setrlimit()` fail.
Detects and reports specific termination signals (like `SIGXCPU` for time limits and `SIGSEGV` for memory limits).